Javascript Libraries + Webpack

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What is library?

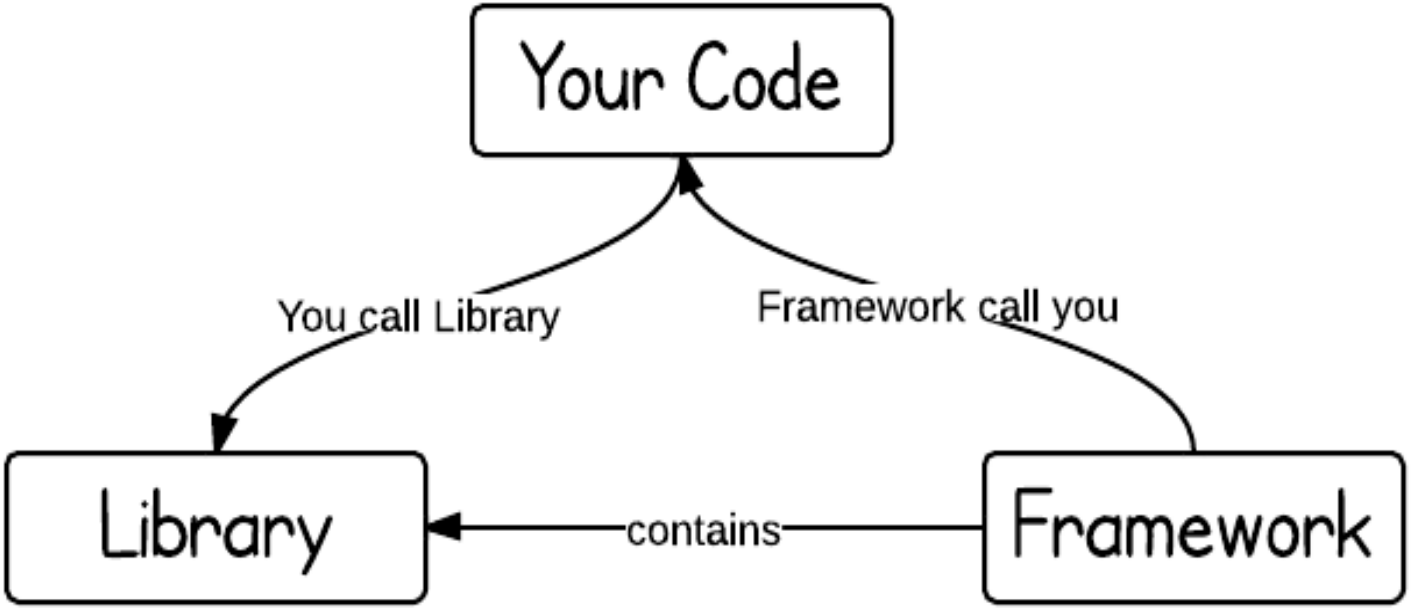
- A JavaScript library is a library of pre-written JavaScript code which allows for easier development of JavaScript-based applications
- An organized collection of useful functionality



Why do we need libs?

- **Saving effort and time**
- **Increasing quality and safety**
- New team members may already be familiar with the library
- Libs are crossbrowser
- Community
- Documentation

Lib vs Framework



Frameworks

- Angular
- Vue
- Svelte
- Next.JS
- Nest.js

…









Tools

- Webpack
- Gulp
- Babel
- ESLint

…









How to add lib to your project?

- NPM/YARN
- HTML tag <script>

```
<script src="https://code.jquery.com/jquery-3.4.1.min.js"></script>
<script src="./js/libs/lodash.js"></script>
<script>
    console.log($);
</script>
```

How to determine if the lib is good?

Install

```
> npm i leaflet
```

Weekly Downloads

390 323



Notifications | Star 30.5k | Fork 4.8k

Leaflet API reference

Map	UI Layers	Other Layers
Usage example	Marker	LayerGroup
Creation	Popup	FeatureGroup
Options	Tooltip	GeoJSON
Events		GridLayer

How to manage your libraries?

Follow semver and update your libs regularly

```
"dependencies": {
  "lodash": "^4.17.21",
  "cors": "~2.8.5",
  "express": "git+https://github.com/zzuhan/express.git"
},
```

Don’t forget to run npm audit fix

```
~/VSCodeWorkspace/nest-js-docker-mongo (master) $ npm audit fix
npm WARN optional SKIPPING OPTIONAL DEPENDENCY: fsevents@2.3.2 (node_modules/fsevents):
npm WARN notsup SKIPPING OPTIONAL DEPENDENCY: Unsupported platform for fsevents@2.3.2:

up to date in 4.385s

70 packages are looking for funding
  run `npm fund` for details

fixed 0 of 0 vulnerabilities in 990 scanned packages
```

Which libraries will you use in your lab



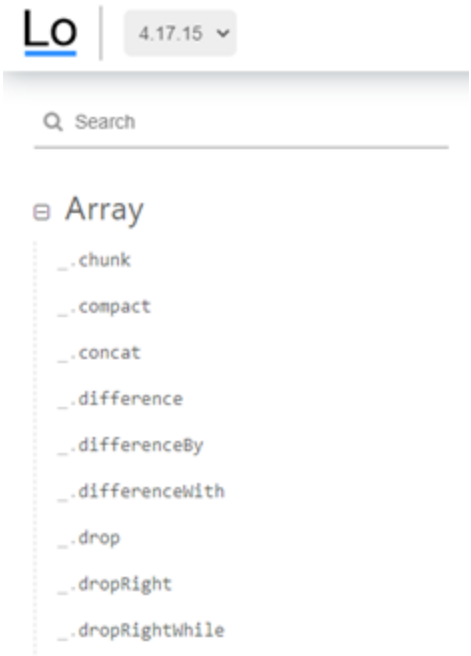


Chart.js





Lodash




```
import { chunk, countBy, sortedUniq } from 'lodash';

chunk(['a', 'b', 'c', 'd'], 2);
// => [['a', 'b'], ['c', 'd']]
sortedUniq([1, 1, 2]);
// => [1, 2]
countBy([6.1, 4.2, 6.3], Math.floor);
// => { '4': 1, '6': 2 }
```

Leaflet

```
<link rel="stylesheet" href="https://unpkg.com/leaflet@1.7.1/dist/leaflet.css"
integrity="sha512-xodZBNTC5n17Xt2atTPuE1HxjVMSvLVW9ocqUKLsCC5CXdbqCmblAshOMAS6/keqq/sMZMZ19scR4PsZChSR7A=="
crossorigin=""/>
<script src="https://unpkg.com/leaflet@1.7.1/dist/leaflet.js"
integrity="sha512-XQoYMqMTK8LvdxXYG3nZ448hOEQiglfqkJs1NOQV44cWnUrBc8PkAOcXy20w0vlaXaVUearIOBhiXZ5V3ynxwA=="
crossorigin=""></script>

...

<div id="mapid"></div>

...

<style>
    #mapid {
        height: 180px;
    }
</style>
```

Leaflet

```
<script>
    L.tileLayer('https://api.mapbox.com/styles/v1/{id}/tiles/{z}/{x}/{y}?access_token={accessToken}', {
        attribution: 'Map data &copy; <a href="https://www.openstreetmap.org/copyright">OpenStreetMap</a> contribute
        maxZoom: 18,
        id: 'mapbox/streets-v11',
        tileSize: 512,
        zoomOffset: -1,
        accessToken: 'your.mapbox.access.token'
    }).addTo(mymap);


    var mymap = L.map('mapid').setView([51.505, -0.09], 13);
</script>
```

Chart.js

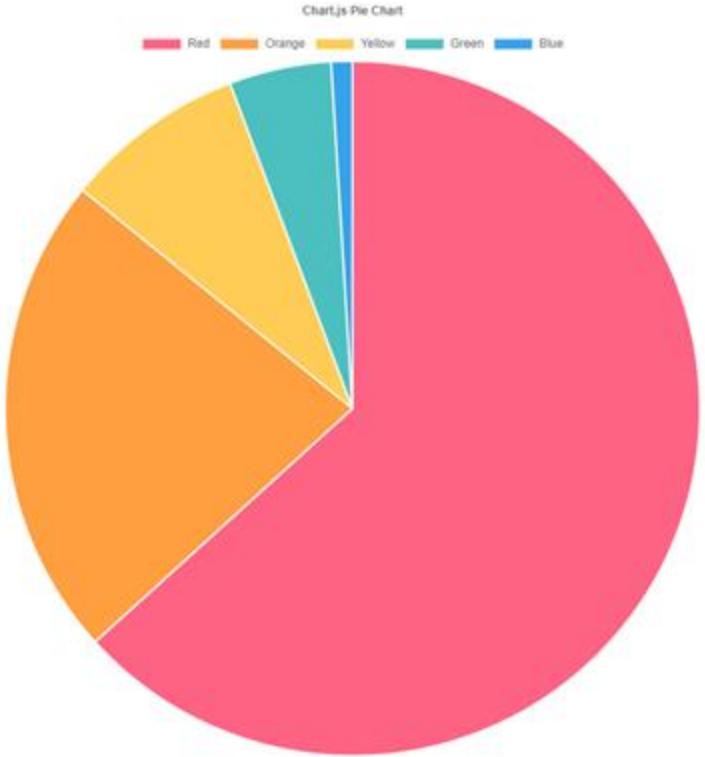

Chart.js Pie Chart
Red
Orange
Yellow
Green
Blue


Chart.js Horizontal Bar Chart
January
February
March
April
May
June
July
-100
-80
-60
-40
-20
0
20
40
60
Dataset 1
Dataset 2

Day.js

- Installation
- Parse
- Get + Set
  - Get + Set
  - Millisecond
  - Second
  - Minute
  - Hour
  - Date of Month
  - Day of Week
  - Day of Week (Locale Aware)
  - ISO Day of Week

```
<!-- CDN example (unpkg) -->
<script src="https://unpkg.com/dayjs@1.8.21/dayjs.min.js"></script>
<script>dayjs().format()</script>
```

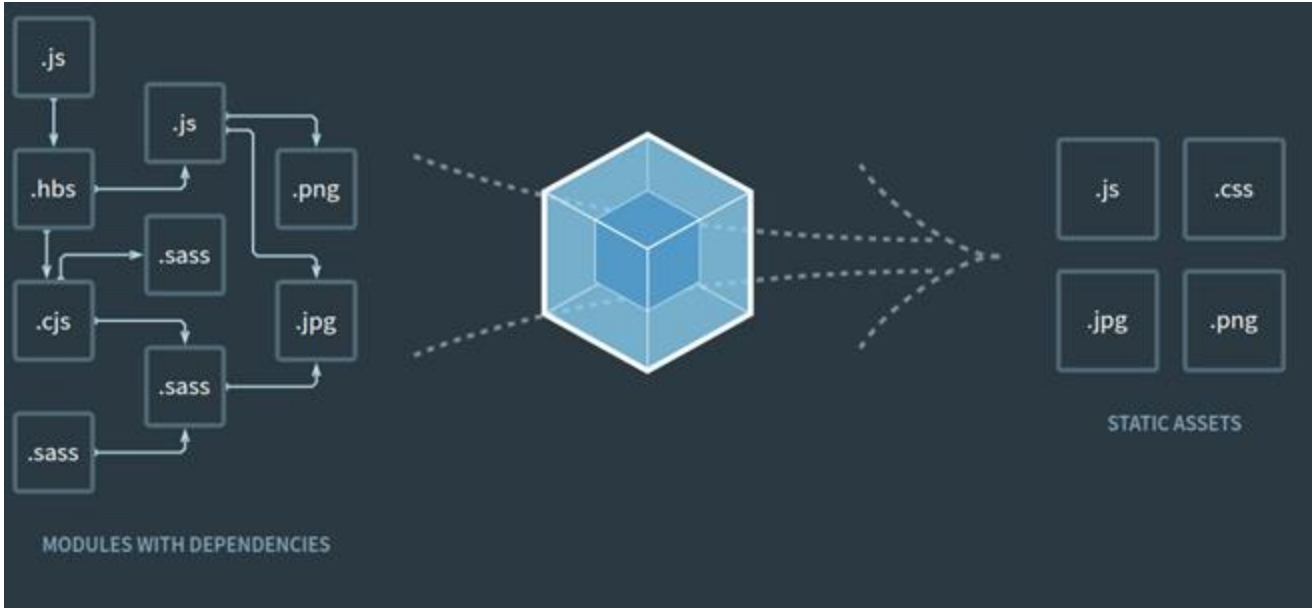
```
dayjs().format()
// current date in ISO8601, without fraction seconds e.g. '2020-04-02T08

dayjs('2019-01-25').format('[YYYYescape] YYYY-MM-DDTHH:mm:ssZ[Z]')
// 'YYYYescape 2019-01-25T00:00:00-02:00Z'

dayjs('2019-01-25').format('DD/MM/YYYY') // '25/01/2019'
```

Webpack

Webpack is a module bundler. Its main purpose is to bundle JavaScript files for usage in a browser, yet it is also capable of transforming, bundling, or packaging just about any resource or asset.



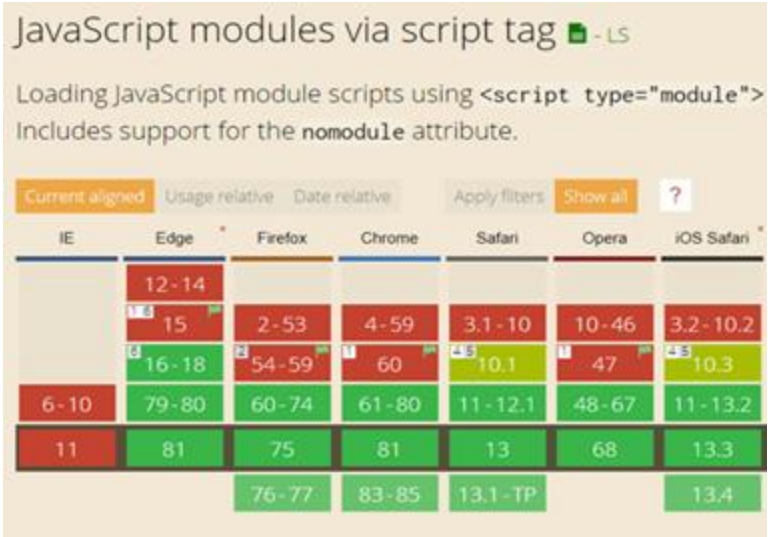
What problem does the bundler solve?

- Modular system (providing js modules for unsupported browsers)
- Allowing to load assets as modules inside your js code
- Adding support for popular loaders and plugins (babel, typescript)

```
import styles from './styles.css';

const divElement = document.createElement('div');
divElement.className = styles['my-class'];
```





Webpack overview

- 59k + stars
- Bundles es6, commonJS and AMD modules
- Wathches the changes and reruns the task
- Embeded dev server, source maps, minification
- A lot of loaders and plugins
- Separate bundles, common chunks

Webpack alternatives





Legacy



Vite



esbuild

An extremely fast JavaScript bundler

Modern

Installation



```
npm install --save-dev webpack
```

```
yarn add webpack --dev
```

Configuration

Default config filename should be: webpack.config.js
Config core concepts:
1. Entry
2. Output
3. Loaders
4. Plugins
5. Mode

Run

1. **With npx (NodeJS 8.2+):**

```
npx webpack --mode development
npx webpack --mode production
```

same as: *./node_modules/.bin/webpack --mode ...*

2. **With npm script (in package.json):**

```
"scripts": {
   "dev": "webpack --mode development",
   "prod": "webpack --mode production"
}
```

```
npm run dev
npm run prod
```



Development and production modes

DEVELOPMENT

PERFORMANCE

2588ms - 42.0 kB

```
> webpack --mode development

Hash: 6ec7e290aa621017ea4f
Version: webpack 4.1.1
Time: 2588ms
Built at: 2018-3-20 17:01:07
    Asset       Size  Chunks             Chunk Names
 main.js   41.7 KiB    main  [emitted]  main
main.css  286 bytes    main  [emitted]  main
```

PRODUCTION

BUNDLE SIZE

3790ms - 25.4 kB

```
> webpack --mode production

Hash: dd8979f922c4fd5e592e
Version: webpack 4.1.1
Time: 3790ms
Built at: 2018-3-20 17:04:07
    Asset       Size  Chunks             Chunk Names
 main.js   25.2 KiB       0  [emitted]  main
main.css  226 bytes       0  [emitted]  main
```

Default config for version 4+

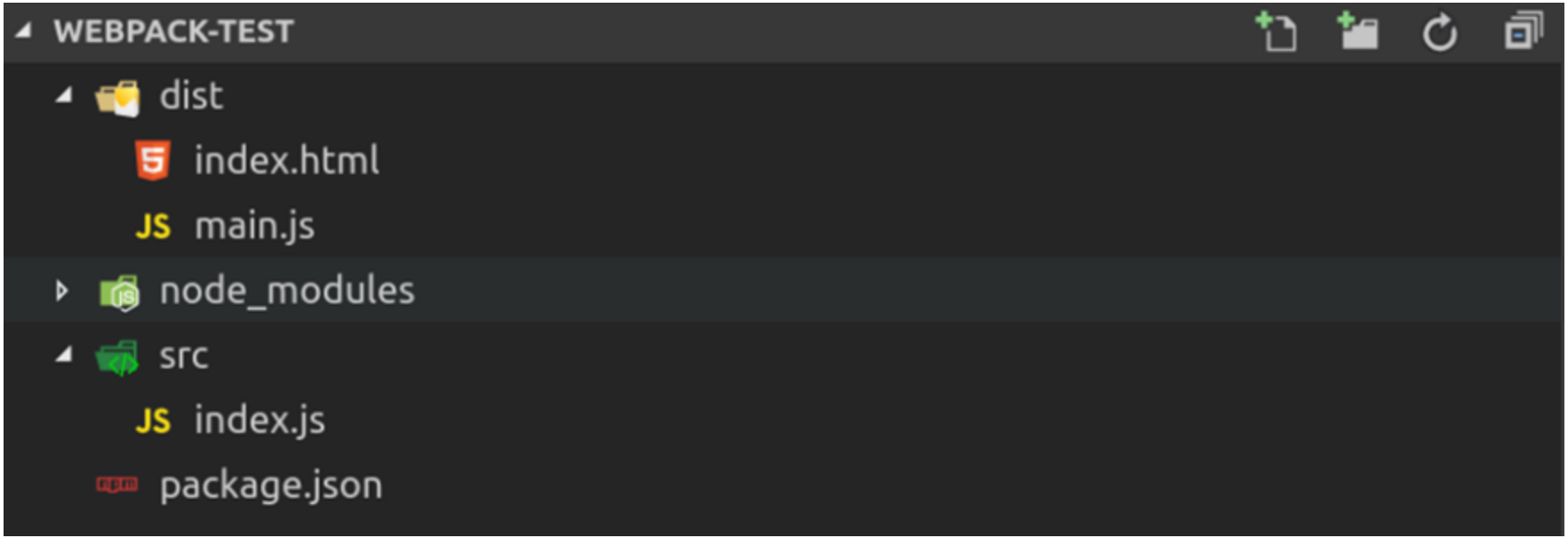
```
const path = require('path');

module.exports = {
  entry: './src/index.js',
  output: {
    path: path.resolve(__dirname, 'dist'),
    filename: 'main.js'
  }
};
```

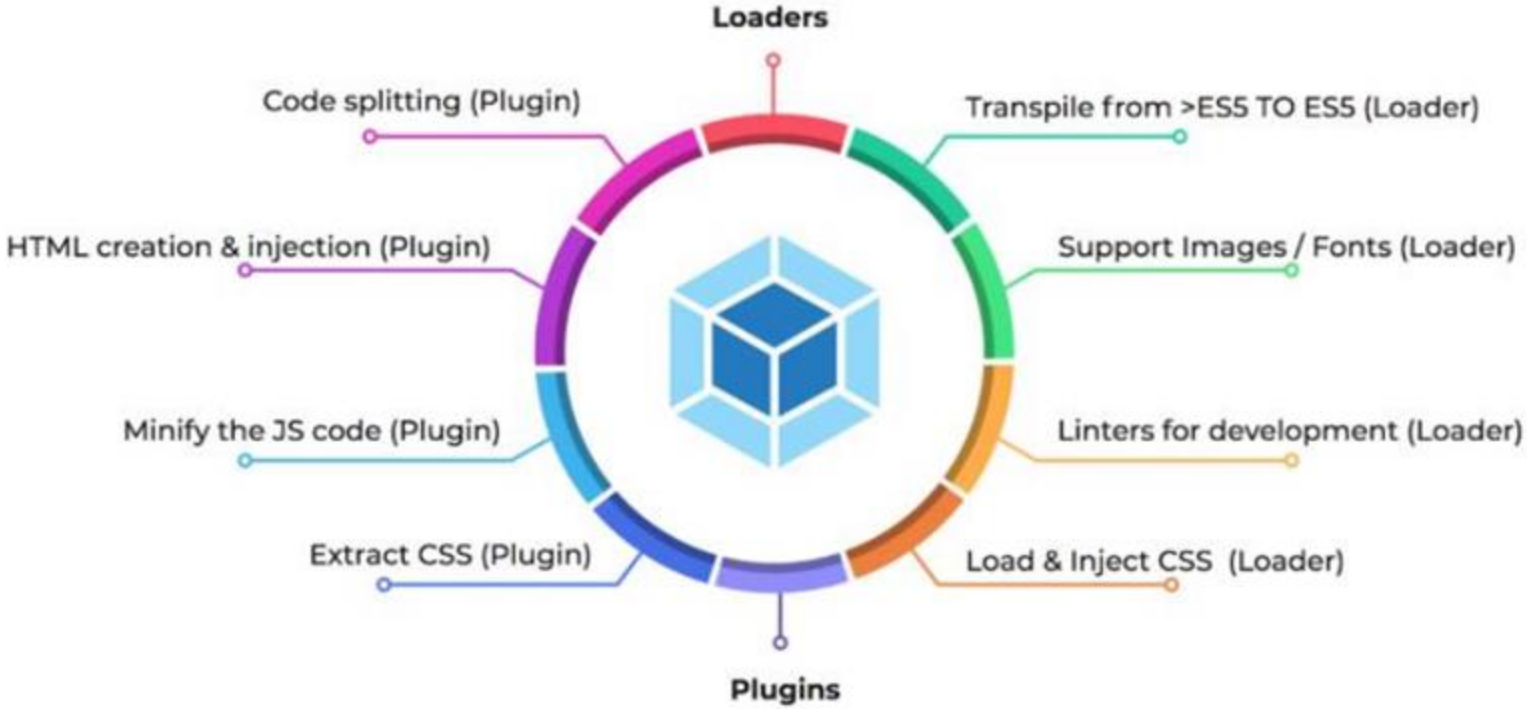
index.html

```
<!doctype html>
<html>
  <head>
    ...
  </head>
  <body>
    ...
    <script src="dist/main.js"></script>
  </body>
</html>
```

File structure



Loaders and plugins



Babel loader

```
module.exports = {
  /*...*/
  module: {
    rules: [
      {
        test: /\.js$/,
        exclude: /(node_modules)/,
        use: {
          loader: 'babel-loader',
          options: {
            presets: ['@babel/preset-env']
          }
        }
      }
    ]
  }
  /*...*/
}
```



Style loader

```
module.exports = {
  /*...*/
  module: {
    rules: [
      {
        test: /\.css$/,
        use:
          [
            'style-loader',
            'css-loader',
          ]
      }
    ]
  }
  /*...*/
}
```



postcss-loader



node-sass
sass-loader

Plugins

```
module.exports = {
  /*...*/
  plugins: [
    new CleanWebpackPlugin(['dist']),
    new HTMLWebpackPlugin({ ... }),
  ]
  /*...*/
}
```

Questions?

Thanks for attention!